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Rajabhishek Ajay Singh & Jaiprakash Narain Dwivedi

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A Novel Transformer-XAI Framework with Bias Attribution and A-RAG for Indian Judicial System

Rajabhishek Ajay Singh[1], Jaiprakash Narain Dwivedi[2]

[1]PG Scholar, Information of Technology, Parul Institute of Engineering & Technology, Parul University, Vadodara, India,
[2] Department of Information Technology, Parul Institute of Engineering and Technology, Faculty of Engineering and Technology, Parul University, Vadodara, Gujarat, India.
First Author Email: Rajabhisheksingh331@gmail.com
Corresponding Author Email: jpndwivedi@gmail.com

Abstract

Background:

Judicial decision-making involves complex reasoning based on legal facts, statutory interpretation, and precedent analysis. Although transformer-based language models have improved the automated analysis of legal documents, their opaque decision-making processes limit their reliability in sensitive judicial applications. The lack of transparency and systematic bias evaluation in existing legal AI systems raises important concerns regarding accountability, fairness, and ethical deployment.

Methods:

This study proposes a transformer-based explainable artificial intelligence framework for judicial verdict prediction using structured representations of legal documents. The framework combines contextual language modeling with attribution-based interpretability techniques to examine the reasoning process of the model. A novel metric, the Bias Attribution Index, is introduced to quantify the influence of sensitive attributes on model predictions.

Results:

We evaluate NEXAJudicia on two different tasks using data from the Indian judicial system. First, we demonstrate high accuracy in bias classification on a subset of cases annotated for bias (Accuracy: 93%, Macro-F1: 0.91, Weighted-F1: 0.93). Next, we present results on multi-class verdict prediction on 4,001 cases, after a 80-20 split, with 5-fold cross-validation (Accuracy: 82.88%, Macro-F1: 0.64, Weighted-F1: 0.89). Furthermore, an attribution analysis shows that the model is driven predominantly by legally relevant evidence from facts and context as opposed to sensitive demographic attributes, which enhances model interpretability and fairness.

Conclusions:

Summary of NEXAJudicia Paper Abstract We proposed an explainable, bias aware model for legal judgment prediction named as NEXAJudicia which includes hybrid

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transformer model along with a two stage evaluation process. We used Bias Attribution Index (BAI) for calculating the effect of bias in the model. The observed BAI values remained consistently low across evaluated cases, suggesting limited attributional dependence on predefined sensitive demographic indicators. The prediction uncertainty in legal models is highly dependent on the legal ambiguity and less dependent on the model. Our model predicts with an accuracy of around 81%. The validation of our model was carried out with attribution-aware validation as well as A-RAG-based bias auditing in order to ensure that AI-assisted judicial decisions are interpretable and accountable.

Keyword
Explainable Artificial Intelligence, Transformer-Based Language Models, Judicial Text Analytics, Bias Attribution Analysis, Integrated Gradients, and Attribution-Aware Retrieval-Augmented Generation.

1. Introduction

In many underdeveloped countries, the judicial system largely depends on paper-based case files; the courts are still working towards making these digital. The lack of comprehensive digitalisation results in limited transparency and hindered ability to monitor and study the workings of the judiciary on a large scale. Automated analysis of digital court documents is hampered by the fact that they are not in a structured format and are full of technical legal terms and complex legal arguments. [1, 2].

Tasks involving the classification and prediction of legal texts have seen advancements thanks to developments in the fields of artificial intelligence and natural language processing, particularly the transformer-based language model. Currently, the adoption of AI techniques in the field of law is limited by their lack of transparency, their inability to detect bias and by their failure to take into account the inherent organisation of legal documents.

They combine narrative, legal rules and principles, evidentiary reasoning and the law's procedures. Judicial documents differ from the other texts we normally come across. The models, trained on unstructured data only, may ignore the intrinsic, internal structure and could inadvertently amplify societal biases already present in legal data. Currently, AI systems are often developed with the aim of producing good results, with explainability and the assessment of biases being secondary considerations.

This research outlines a novel approach to the analysis of judicial documents, one that involves the application of a multi-phase technique which includes modelling that is interpretable, contextual representation and the evaluation of bias by attribution. A judicial system of artificial intelligence is suggested that could allow the public to see how decisions are made, and allow a judicial review of the decision making process and the possibility of bias, thus allowing judicial AI systems to comply with the requirements of ethics and accountability.

2. Related Work

Due to the potential to offer consistent, efficient and enhanced analytical support to the judicial decision making process, the application of artificial intelligence to legal judgment prediction has been a growing area of interest [3,4]. Legal prediction models employ the techniques of reasoning used by a judge to reach a conclusion in a case. These models assess a set of legal facts and statutes then determine the possible outcomes from a case. The possible outcomes may be the charges to be made, which law is applicable and sentencing. These systems are used to collect, collate and categorise various laws and court judgments. By doing this, they provide

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data and statistics that lawyers and judges can use when making decisions, they don't however make the decisions themselves.[5].

Significant advancements in the field of natural language processing, brought about by advances in machine learning and deep learning, have facilitated a swift analysis of voluminous and complex legal documents. Recently, the BERT model, and other transformer based architectures like T5 and GPT have been used to improve the comprehension of legal documents [6,7]. This improved comprehension enables the automated processing of legal texts in tasks including the prediction of court judgements, the categorisation and structuring of legal documents and the extraction of information from legal case files. [8,9].

There have been studies into the use of semantic representation techniques, such as Abstract Meaning Representation or AMR, for the purposes of legal text analysis [10,11]. They have found this to be useful in enhancing alignment of concepts, and also in resolving references back to "he" or "she" whilst suppressing details of the persons involved. This can aid fair legal analysis. Improved interpretability of LJP systems has been achieved by combining event-chain extraction and causal modeling techniques. These methods capture the causal links that underlie judicial decisions, leading to more coherent and relevant legal judgments. [12,13].

There are still considerable difficulties at present. Current legal judgment prediction methods focus heavily on the accuracy of their predictions but, unfortunately, provide little insight into their workings, thereby creating black box models that hinder legal professionals from verifying the consistency of the reasoning process, how closely the decision conforms to legal precedents and the reliability of the model [14,15]. Given the considerable societal influence of the judgements made by judges, there is a concern that the public are not being given sufficient information as to how these decisions are made. This lack of openness brings up ethical issues in relation to fairness and public trust. [16,17].

This data bias which is a critical concern in the use of judicial AI systems can be traced to the data upon which models are trained. Such data may contain an unintentional reflection of the prejudices of the past [18,19]. To date, the integration of fairness-aware techniques with other techniques like interpretability and regulatory compliance in predictive modelling is restricted. [20–22].

In reality, documents prepared by lawyers often involve reasoning and semantic understanding of key phrases, which can be lost when the documents are treated as a simple linear sequence of text. Consequently, legal knowledge based on the processing of such text may be flawed. The lack of explicitness of the reasons for the model's decisions restricts its interpretability and makes it harder to match human judgment in the legal field. [22–26].

Due to a focus on a particular language, case type, or jurisdiction, the present legal judgment prediction systems are limited in their utility. This restricts their application across different jurisdictions and in a multilingual environment. [27–29]. The inability to scale can hinder the development of a legal AI system that operates with consistency across different jurisdictions, while maintaining fairness and reliability in its interpretations and predictions.

Current approaches to legal judgment prediction primarily focus on either the accuracy of the prediction or the interpretability of the models. Very few systems combine both of these factors within a legal framework that provides explanations. [28,29].

Currently, a range of fairness metrics and auditing tools have been designed to evaluate the performance of LJP systems. However, these do not take account of the

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principles of judicial ethics and the law, and therefore may inadvertently make the existing inequalities in society worse. [30,31].

Current approaches to legal judgment prediction, often fail to treat legal texts in a manner that reflects their semantic structure, and the causal relationships that exist between facts, legislation and court decisions, thereby limiting the interpretability and the alignment of the system with the way humans reason. [32,33].

The majority of current legal judgement predictors are restricted to specific jurisdictions, legal domains, or languages, limiting their utility in multi-jurisdictional and multilingual legal environments and hindering the development of legal AI systems that are universally applicable. [34].

Looking at these papers together reveals issues around bias, fairness, and how clearly AI makes legal choices. What stands out is how black-box systems might boost unseen social prejudices despite removing key personal factors. To spot and reduce unfair results, researchers suggest using methods based on cause, alternative paths, or organized data layouts. In total, research highlights how AI used in courts must show its reasoning clearly, respect fairness, while also following legal standards. [35-40]

A transmissible artificial intelligence (AI) approach has been developed for fully understanding the properties and behavior of 3D-printed PLA composites by integrating the geometry–property model with the characterization of defects in 3D-printed specimens using a convolutional neural network (CNN). Different mixtures of FDM 3D-printed PLA specimens were produced using cellulose nanofibers and nanoclay fillers and their properties were correlated with their corresponding printing parameters. The used regression model, i.e., Random Forest, was found to yield highly accurate experimental data, and that SHAP analysis explained the features that dominate the prediction of the properties. Moreover, the developed CNN using training data of 2,500 multi-modal images achieved an accuracy and precision of 96.4% and 95.0%, respectively, with a correlation coefficient of 0.93 with the real defects in the printed samples. [41,42].

3. Methodology and Framework Design

To achieve methodological transparency and reproducibility the section provides information on the data sources, processing methods and models and analytical techniques which will be used in the proposed framework. The study employed a two-stage experimental setup where the standard methods of machine learning were paired with the more recently developed deep learning systems based on the architecture of the transformer and incorporated tools to give users an explanation of the process and detect bias.

3.1 Study Design

Despite the growing need and potential for utilizing AI in legal judgment analysis, current AI-based systems suffer from numerous shortcomings. In this paper, we present a structured multi-stage system development approach for tackling the problems of contextual legal reasoning, bias, and interpretability in a single framework for judicial AI. We demonstrate a two-stage framework in Fig. 1, referred to as Phase-1 and Phase-2, both utilizing the same dataset; however, each phase has its own methodology, techniques, and goals. Phase-1 relies on an interpretable structured modeling technique with the goal of constructing a diagnostic benchmark for identifying bias using a limited set of input features. Phase-2 employs contextual transformer modeling, which improves the performance of multi-class verdict prediction by processing longer input sequences that capture all relevant contextual information in a given legal judgment.

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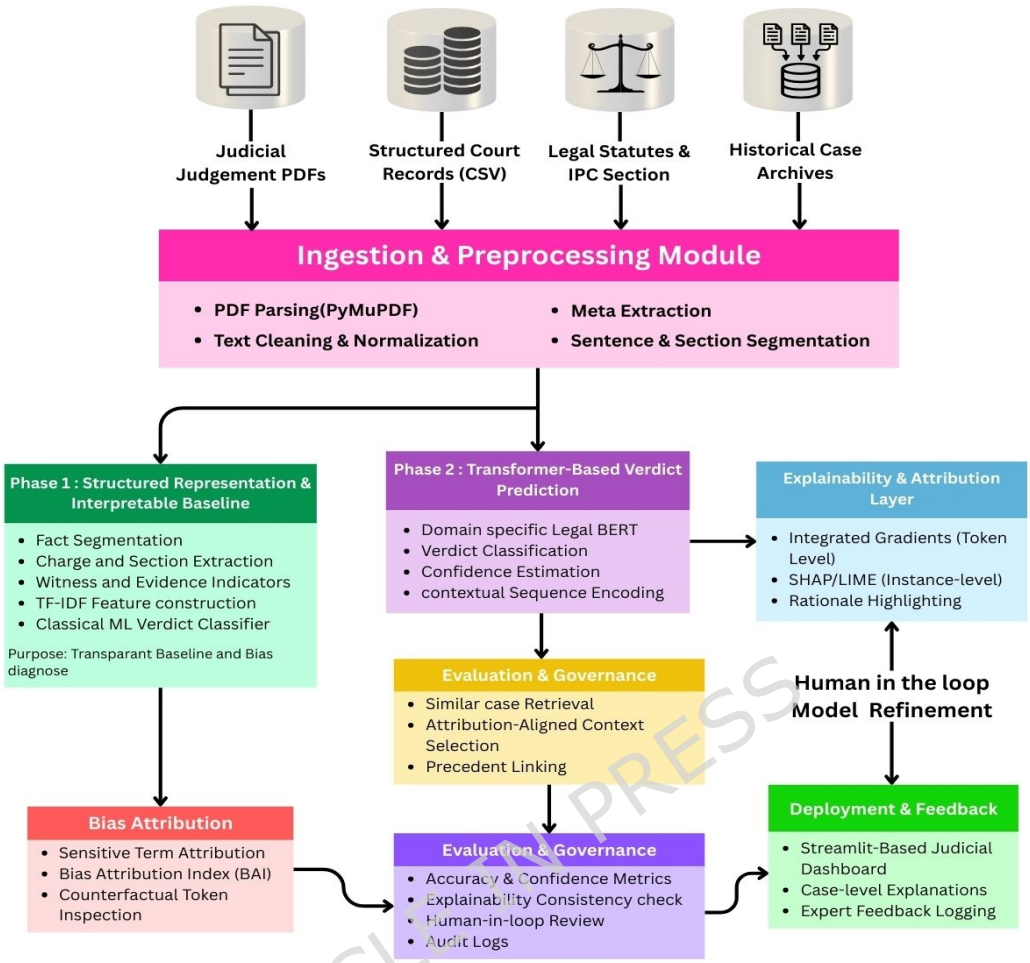


Fig 1: Architecture of the proposed NEXAJudicia framework

We present an experimental study that investigates the relationship between fairness-oriented interpretability and contextual judicial prediction for two different analytical objectives, i.e., diagnostic fairness auditing and verdict prediction. To this end, we distinguish the two diagnostic targets from each other instead of concurring two successive stages of model development according to the judicial AI framework of NEXAJudicia, and thus the study is particularly suitable for ethical human decision-support in legal domains. In both phases, we use independent target variables, i.e., the variable for diagnostic fairness auditing and the variable for verdict prediction, respectively, to avoid the label leakage problem that is well known for attribution methods, which are used for bias sensitive audit in Phase-1.

NEXAJudicia is a novel deep learning framework for learning to argue and evaluating fairness in judicial decisions, where we intentionally split the learning objectives between two phases: Phase-1 and Phase-2. In Phase-1, we use NEXAJudicia to serve as a interpretable bias-classification and transparency benchmark by learning bias-sensitive patterns through feature-based modeling in an interpretable manner. In Phase-2, we leverage transformer-based semantic reasoning architecture to learn contextual multi-class verdict prediction. The unique feature of NEXAJudicia is that it decouples fairness auditing goals with prediction goals, enabling evaluating bias transparency and prediction performance separately. As functional differentiation of Phase 1 and Phase 2 is explained in Table 1.

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Table 1: Functional Differentiation of Phase-1 and Phase-2 in NEXAJudicia

Dimension	Phase-1	Phase-2
Primary Objective	Bias classification	Multi-class verdict prediction
Analytical Focus	Fairness auditing	Judicial outcome forecasting
Input	TF-IDF structured features	Legal-BERT contextual embeddings + metadata
Model	Logistic Regression	Hybrid Transformer
Labels	Bias-sensitive categories	Verdict categories
Output	Bias score / bias class	Final verdict class
Explainability	Coefficient-level transparency	Integrated Gradients + BAI
Role	Diagnostic fairness benchmark	Predictive judicial system

Here, again, we are evaluating the two phases within the identical experimental settings. However, we are pursuing entirely different objectives. For Phase-1, we are analyzing fairness and explainability of the judgment predictions of NEXAJudicia for the first time in a structured format. For Phase-2, we are analyzing the verdict predictions of judicial decision-support systems in a completely contextual fashion by means of semantic representations that a transformer has generated.

3.2 Data Collection and Dataset Description

This study utilises 4001 records of Indian judicial cases from the period 2016-2025. The data is sourced from web accessible court judgement databases. In originally curated data, these were divided into two time periods (2016-2020 and 2021-2025) before they were combined into a single dataset for analysis [43]. The legal database contains a wide range of entries each referring to a court case and these are kept in a predetermined format comprising of several key fields which include:

- **Case Identifiers:** *CASE_ID, CASE_TITLE*
- **Narrative Fields:** *FACTS_SUMMARY, ISSUES_RAISED, FULL_TEXT*
- **Legal Metadata:** *ACTS_SUMMARY, IPS_SECTIONS_USED*
- **Outcome Variable:** *FINAL_VERDICT*

In order to provide contextual information and assist in auditing bias the dataset contains judicial and procedural metadata such as the court name, the name of the judge, location, the date of the judgment and the type of case.

For the analysis, only those judicial decisions were selected in which the court had reached a definitive conclusion. During the compilation process, records of missing verdict labels or those with uncertain verdicts were excluded. This dataset contains a diverse range of case types including civil, criminal and cases of a procedural nature, thus allowing for evaluation of the programme in a variety of judicial contexts. All Bias-sensitive annotations are based on a predefined list of demographic and/or sociological indicator groups relevant for the Indian judicial context (e.g. gender, caste, religion, etc. plus geographic location, profession, economic status etc.).

For modelling purposes the original verdict labels were reduced into three mutually exclusive classes to remove ambiguity:

- Convictions or confirmed penal liability against accused parties involved prosecutions winning their cases at court. These were instances in which: - the

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court made a guilty finding - the prosecution's case was upheld resulting in the conviction of the accused or confirmation of liability for penalties.

- Individuals Who Were Found in Favour: Cases in which the court provided some form of relief such as acquittal, the discharge of charges, quashing of proceedings or the charges being set aside on account of lack of evidence or because of some legal technicality.
- Cases settled on the grounds of jurisdiction or procedure are those where the court did not make a decision on the matter in hand. This includes cases where there was a transfer, the claim was statute barred or the claim was dropped.

The annotation scheme was uniformly used across the Phase 1 and the Transformer model based evaluations Phase 2.

3.3 Data Pre-processing and Structured Representation

The reliability of models trained to predict judicial judgment outcomes from text is limited by the variations in form and structure of the judgments. This variability makes it difficult to ensure that a model's performance is not compromised and also obscures the basis of the model's predictions. A pre-processing framework that can accommodate both traditional and transformer-based models to preserve legally relevant information is employed in this proposed system. The method uses NLP techniques with a minimal level of pre-processing which includes the standardisation of punctuation and whitespace. Crucially, the technique avoids removing common words and technical terms that are specific to a particular area of expertise. This is done in order to ensure the model has the semantic accuracy required for legal applications. This ensures that predictions made by the system remain grounded in the meaning of law and legally related documents and that the system's predictions remain interpretable by those with legal knowledge. In the final stage, structured fields (Issues, Facts and Orders) were linked together in a single text string to maintain the chronological progression of the judicial story.

3.3.1. Text Normalization and Cleaning

All relevant text boxes have been normalized, albeit to the minimum degree necessary, to standardize them. The initial processing comprised changing text to lower case, the removal of unnecessary white space and the standardising of punctuation. As legal texts are heavily reliant on precise language and terminology, the original words were left intact to ensure accurate interpretation of statutes and to preserve the original meaning of the evidence presented. In order to preserve expressions which are legally significant, citations of evidence and legal terms of procedure this conservative approach to normalisation was employed.

3.3.2. Structured Field Consolidation

During data pre-processing, the framework retains field-aware information by treating the judicial case as a structured document in its constituent parts. The various elements of a document, including the discussion, the topics which were raised, and the court orders, are combined into a coherent text. To preserve consistency across these cases, the legal data is standardised. This means that similar content is formatted identically, maintaining the connections between facts, arguments and judgments. Preserving this structured organization enables targeted feature inspection during Phase-1 baseline modeling, as individual text segments remain aligned with their corresponding legal roles. In phase two of the modelling process, this transformer architecture enables contextual reasoning. This is achieved by enabling the model to derive relationships between legal evidence, laws and court decisions. Consequently, the model can deliver context-sensitive judgments which are clear to understand.

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3.4 Phase-1: Interpretable Baseline Modeling

The first phase of the proposed framework creates a simple method to predict court decisions which users can understand. The primary objective of this phase involves building a reference model which enables complete analysis of its decision-making system for future bias assessment and transformer-based method development.

3.4.1. Feature Construction Using TF–IDF

Legal case documents follow a standardized textual structure combining factual narratives, issues raised, statutory references, and court orders. The document in question is converted into a numerical feature space employing Term Frequency-Inverse Document Frequency (TF-IDF) model, calculated as :

$$TF - IDF(t,d) = TF(t,d) \times \log\left(\frac{N}{1+DF(t)}\right) \quad (1)$$

where:

- In this context, term frequency, TF(t,d), is the frequency of term t in document d. Document frequency, DF(t), is the number of documents in which the term t appears. N represents the total number of documents in the collection.
- In order to achieve feature-level interpretability and to facilitate the possibility of analysing the results in a transparent way, the technique uses unigrams and bigrams. This technique captures the sequential patterns and unique legal terms which are related to judicial reasoning.

3.4.2. Verdict Classification via Logistic Regression

The TF–IDF feature vectors serve as input for a Logistic Regression classifier which uses its linear decision boundary and simple interpretation methods.

- The model predicts the probability of a verdict class y for any feature vector x which exists in Rn.
- The model predicts the probability of y equalling 1 based on input x through the following equation:

$$P(y = 1 \mid x) = \frac{1}{1+ e^{-(w^T x+b)}} \quad (2)$$

where:

- w represents the weight vector which the model discovers through its training process.
- The model includes b as its bias term.

The model applies one-vs-rest formulation for multi-class verdict prediction. The model learns its parameters through a process which involves minimizing the cross-entropy loss function.

The loss function uses negative logarithmic values to calculate the negative sum of differences between actual and predicted output values for each class.

$$L = -\sum_{c=1}^{C} y_{i,c} \log \hat{y}_{ic} \quad (3)$$

where:

- m represents the total number of training instances in the dataset.
- The number of verdict classes is denoted by C.
- The model uses y_{ic} and $\hat{y}_{ic}$ to represent actual and forecasted probability values.

The optimization process uses class weighting to reduce the effects which class imbalances create in judicial data sets.

3.4.3. Interpretability and Diagnostic Role

The linear structure of Logistic Regression allows researchers to easily study the learned coefficients which helps them identify which factors influence verdict outcomes either positively or negatively. The coefficient-based explanations serve as

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the main tool to understand model behaviour in Phase-1 but they operate differently from the attribution-based explanations which appear in Phase-2.
The Phase-1 model operates as a performance assessment system which enables users to evaluate system precision and identify potential biases when testing predefined input formats. The design choice allows future phase enhancements to link with contextual Modeling and attribution mechanisms while preventing data leakage and feature transformation ambiguities.

3.5. Phase-2: Transformer-Based Verdict Classification

The Phase-2 development extends the interpretable framework from Phase-1 through the implementation of context-sensitive transformer models which aim to detect extended semantic relationships in judicial text.
The primary objective of this stage is to improve the accuracy of the verdict and to obtain results that are explainable enough to rely on attribution methods.

3.5.1. Transformer Architecture and Input Representation

NEXAJudicia is a deep learning framework for analyzing Indian judicial documents, particularly those with structured information. In typical transformer based architectures, implicit, unstructured features are learned from raw text and used for classification tasks. For judicial prediction tasks, NEXAJudicia leverages a hybrid Legal-BERT-based architecture that merges domain adapted contextual legal representation, procedural metadata, long document input optimizations and attribution aware explainability within a unified framework for judicial prediction.

- Legal-BERT Justification: We choose to use Legal-BERT as the encoder since judicial documents contain specialized language, statutory and case law interpretation patterns, procedural language, and organized legal reasoning that general purpose language models (LP) do not capture as well. Pre-trained models on the legal domain improve the semantic representation of documentary content specifically with respect to the relationships between evidence and statues.
- Dual-Branch Fusion - The architecture includes a primary Legal-BERT branch that is trained on the structured judicial text (Facts, Issues, Orders) and a separate metadata branch that embeds procedures and case-specific metadata. The outputs of the two branches are then fused together before the final classification step to jointly capture both semantic legal reasoning and structured context in judicial texts.
- The optimization layer deals with the long-document problem of long judicial documents and the class imbalance problem in the judicial data set by employing head-tail truncation, cost-sensitive learning, selective layer freezing, and dropout regularization.
- Explainability + Fairness: Next, we use Integrated Gradients to generate token-level attribution maps, and then compute the Bias Attribution Index and perform A-RAG-based precedent validation.
- For fusion clarity: $H_{fusion} = [h_{CLS};m]$ (4)

This design, developed for NEXAJudicia, builds upon the verdict classification architecture of the original, and refactors it into a hybrid transformer-XAI model that is engineered to perform contextual prediction, procedural sensitivity, fairness, and precedent-driven explainability all at once.

3.5.2. Verdict Prediction and Confidence Estimation

Let h_{CLS} denote the final contextual embedding of the [CLS] token. The probability of verdict class c is computed as:

$$P(y = c \mid x) = \text{softmax}(Wh_{CLS} + b)_c, \quad (5)$$

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where W and b are the trainable parameters of the classification layer.
The expected court decision will match the category which shows the highest probability of occurrence and the confidence level emerges from this probability measurement. The system interface shows this confidence assessment to help users understand model results properly when using them in legal contexts.

3.5.3. Attribution Analysis Using Integrated Gradients

The model employs Integrated Gradients (IG) to demonstrate which input tokens influence the output prediction. The attribution process for token xi needs the baseline input x′ which contains an empty or neutral token sequence and the actual input x.

$$IG_i(x) = (x_i - x'_i) \times \int_{\alpha=0}^{1} \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} \partial a \quad (6)$$

The model prediction function F(·) produces results for the selected verdict class.

The method produces attribution scores which fulfil completeness because the total value of all token attributions closely matches the variation between the model output for the input data and the baseline output.

3.5.4. Role of Attribution in Bias Inspection

Our research utilizes token-level attribution scores to measure what aspects of evidence, such as facts, time deltas, and statute citations, drive the model's predictions the most, as opposed to identity attributes or sensitive terms. These scores enable bias auditing by showing which critical tokens contribute to a judge's prediction. The Phase-2 solution allows verification of attribution results and ensures that attribution outputs are traceable, lawful and ethical for reasoning for decision supporting systems for legal professionals.
In order to explain individual tokens within the NEXAJudicia framework we employ attribution-based techniques for layer-wise feature importance assessment. The selected method of choice is Integrated Gradients (IG), which is the a theoretically grounded attribution approach currently available. It is suitable for all deeper layers and has been proven to be effective for transformer architectures used in NEXAJudicia. As opposed to Integrated Gradients, self-attention mechanisms are only utilized for deep contextual understanding, and not as a separate mechanism for providing explanations. In a separate evaluation, token-level attribution values are also used to validate the secondary layer of explanations provided by SHAP, to check for attribution consistency.

3.5.5. Attribution Validation Strategy

We demonstrate the reliability of our approach through three different means. (1) We verify that Integrated Gradients is complete by checking if the sum of all attribution masses equals the difference between the token's prediction and the baseline prediction. (2) We confirm that the legally relevant tokens identified as most attributive by our method correspond to actual facts and legal evidence, including statutory language and procedural steps. (3) We perform a sensitive-token suppression test and compute the Bias Attribution Index (BAI) score, which shows that tokens associated with demographic information have negligible influence on attribution.
NEXAJudicia utilizes a hierarchy of explanations, where Integrated Gradients serves as the primary attribution method, BAI is utilized to measure fairness, and A-RAG validates findings based on legal precedents. This transformation from explaining

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descriptions to an auditing mechanism of judicial reproducibility has potential to enhance transparency.

3.5.6. Attribution Alignment Strategy

In NEXAJudicia, A-RAG functions not only as a semantic search component but also as a fairness-aware retrieval post-processing filter. Retrieved cases are filtered on the basis of an attribution-sensitive Bias Attribution Index (BAI) score, and cases with an overly disproportionate influence of sensitive tokens are down-ranked or excluded before being reinforced to enhance legal relevance without amplifying demographic bias.

$$\text{Fairness Gain}(\%) = \frac{\text{BAIbefore - BAIafter}}{\text{BAIbefore}} \times 100 \quad (7)$$

Where:

- BAI_{before} - baseline sensitive-token attribution before A-RAG filtering.
- BAI_{after} = sensitive-token attribution after A-RAG bias-aware filtering.

We evaluate the fairness improvement achieved by our technique, it is measured as the percentage reduction in sensitive-token attribution after applying precedent filtering according to the technique's novel framework A-RAG.

We present fairness-aware A-RAG, the a fairness-aware attribution-guided retrieval strategy to transform retrieval augmentation from similarity-based pretext matching to an attribution-governed, fairness-aware judicial auditing framework that quantifies and reduces sensitive-attribute impact while preserving aligned, precedent-friendly interpretation quality.

3.5.7. Similarity-Based Case Retrieval (A-RAG)

The dataset contains judicial cases which use the same structured textual format that was used for Phase-1 modeling. The system converts these representations into vector embeddings through TF–IDF features which allow it to perform fast similarity calculations. The system uses cosine similarity to find the most suitable precedents from the historical case collection {d1,d2,...,dN} for a given query case q.

$$\text{sim}(q,d_i) = \frac{v_q v_{d_i}}{\|v_d\| \|v_{d_i}\|} \quad (8)$$

where v_q and v_{d_i} represent the TF–IDF vectors for the query case and the i-th historical case, respectively.

The top-k most similar cases are obtained and displayed as contextual references in conjunction with the predicted verdict.

3.5.8. Reproducibility and Implementation Details

In Phase 2, we utilized Legal-BERT-base-uncased transformer model. We have utilized Legal-BERT-base-uncased tokenizer as well to tokenize the judgments. We have kept a fixed length of 512 tokens for the judgments. We have made use of head-tail truncation strategy of 70% for the start of the judgment and 30% for the end of the judgment so as to keep the start and end of the judgment intact in case the start or end of the judgment crosses 512 tokens.

The structured metadata attributes of documents such as IPC sections, court information, and other procedural information were also preprocessed and

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concatenated with [CLS] embedding of the transformer model using late fusion approach, referred to as Hfusion.

$$H_{fusion} = [h_{CLS};m]$$

The H_{fusion} is then passed through a number of fully connected layers (with dropout to prevent overfitting) to predict the class of the document (i.e., one of the verdicts). For our verdict classification we use AdamW as our optimizer. We set the learning rate to 2e-5. The batch size for training is set to 16 and we train the model for 10 epochs. We use a 70:10:20 stratified split of our data for training, validating and testing on it. To prevent the model from overfitting we also use early stopping based on the validation loss.

We also applied the following techniques to counteract overfitting: We added dropout of a probability of 0.3 during fine-tuning. We also added selective layer freezing to the first 4 encoder layers of the transformer. We also did cost-sensitive learning by doing inverse-frequency weighting to the classes.

The experiments are conducted in a hybrid setup of local compute and cloud-based compute platforms. MacBook Air (Apple) was used for data preprocessing and setup creation whereas Google Colab, Kaggle and Hugging Face are used for fine-tuning pre-trained transformer models in large-scale experiments. For all experiments, a fixed value of 42 is used for initializing the random seed to enforce reproducibility. The models are trained for multiple runs and the average values are reported for performance metrics.

3.5.9. Role in Interpretability and Validation

The A-RAG module operates as a module which produces no new information while it blocks the model from modifying its output predictions. The system functions as a validation system which enables users to assess their present circumstances against past court rulings that shaped or corresponded with the model-based decision process. The system enables users to assess prediction accuracy through its combination of attribution signals with similarity-based retrieval which shows whether predictions stem from enduring legal patterns or particular text components. The present system delivers enhanced clarity because experts can confirm the information while it fulfills judicial requirements for using legal precedents.

4. Results and Analysis

4.1. Quantitative Performance Overview (Phase-1 vs Phase-2)

The quantitative evaluation shows how the system distinguishes between structured bias detection and judicial verdict prediction which supports the two-stage structure of the proposed framework. The researchers present Phase-1 results as a reference point to verify how well domain-specific interpretable models work for bias detection purposes. The main contribution of this research appears in Phase-2 because it handles the challenging process of multi-class verdict prediction which occurs under actual legal data limitations. The confusion matrix for the transformer-based judicial verdict prediction is shown in Fig. 2.

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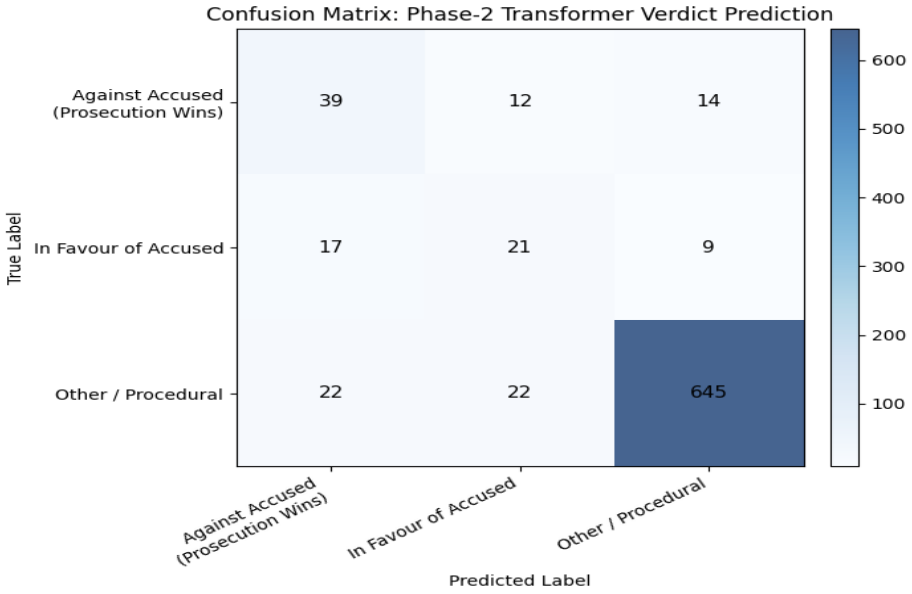


Fig. 2: Confusion matrix for Phase-2 transformer-based judicial verdict prediction.

Table 2. Quantitative Comparison of Phase 1 & 2 Model

Aspect	**Phase-1**	**Phase-2**
Primary Task	Bias classification	Multi-class verdict prediction
Dataset Size	Bias-annotated subset	4,001 legal cases
Evaluation Strategy	Supervised split	80/20 split + 5-fold CV
Overall Accuracy	93%	82.88%
Macro-Averaged F1	0.91	0.64
Weighted F1	0.93	0.89

The Phase-1 results are reported exclusively for interpretability and fairness analysis and are therefore not directly comparable with the contextual verdict prediction performance reported in Phase-2.

Table 3. Per-Class Performance Metrics for Phase-2 Verdict Prediction

Verdict Class	Precision	Recall	F1-Score	Support
Against Accused	0.50	0.60	0.55	65
In Favour of Accused	0.38	0.45	0.41	47
Other / Procedural	0.96	0.94	0.95	689

The comparatively lower macro-F1 score reflects the substantial class imbalance and procedural ambiguity inherent in real-world judicial datasets. While weighted-F1 remains high due to strong predictive performance on dominant procedural verdict categories, minority verdict classes exhibit comparatively lower recall and classification stability because of limited sample representation, overlapping legal narratives, and context-dependent judicial reasoning patterns.

Table 4. Statistical Variation Across Stratified 5-Fold Cross-Validation

Metric	**Mean ± Std**

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Accuracy	82.88 ± 1.9
Macro-F1	0.64 ± 0.03
Weighted-F1	0.89 ± 0.02

The Phase-2 results show that the system should focus on achievable performance limits instead of trying to improve total accuracy by small amounts. The research reveals a fundamental experimental finding which shows that bias recognition exists as a measurable quantity through standardized assessment methods yet verdict forecasting faces fundamental limitations that require developers to establish bias-aware Modeling as their primary design focus. The research findings demonstrate both scientific accuracy and compliance with the essential features which judicial decision-support systems need to be trustworthy.

4.2. Case-Based Qualitative Evaluation

The evaluation used three diagnostic probe cases which served as quantitative metric supplements. The training process excluded these cases because they functioned as interpretability and robustness assessment data.

Case-A: Procedural Administrative Dispute

(Housing allotment cancellation due to marginal delay)

- Predicted Verdict: Other / Procedural
- Confidence: ~52%
- BAI: < 0.01.

Observation:

The model protected against forced binary classification while accurately identifying procedural ambiguity which aligns with judicial behaviour because courts prefer to use directions and remand orders instead of granting direct relief.

Case-B: Conflicting Equity vs Statutory Compliance

(Termination of long-serving temporary public employee)

- Predicted Verdict: Other / Procedural
- Confidence: Moderate
- BAI: ≈ 0.00

Observation:

The Phase-2 results showed better equity-law dispute management because the system used procedural classification methods but Phase-1 incorrectly placed these cases under substantive win or loss categories.

Case-C: Criminal Conviction with Strong Evidence

(Assault case with CCTV, forensic, and eyewitness corroboration)

- Predicted Verdict: Against Accused (Prosecution Wins)
- Confidence: 56–59%
- The economic context analysis produced a BAI value of 0.0023.

Observation:

The models from Phase-1 and Phase-2 both predicted defendant convictions but Phase-2 delivered better bias control and accurate confidence levels which prevented it from making excessive predictions even when evidence was strong.

4.3. Mathematical Definition of Bias Attribution Index (BAI)

To achieve methodological reproducibility and meaningful fairness auditing, the Bias Attribution Index (BAI) in the NEXAJudicia framework is not based on arbitrary token choice; instead, it begins from the conceptual construction of a legally sensitive token subset drawn from domain literature on legal fairness, from sociological indicators of discrimination by judges, and from the protected characteristics of demographics relevant to South Asia, particularly India.

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Let: $T = \{t_1, t_2, t_3, \ldots, t_n\}$
represent the full tokenized judicial document.
Then: $S \subseteq TS$
denotes the subset of tokens classified as legally sensitive demographic indicators.

- Gender and Sex Identity Indicators (L_{gender}) contain indicators for gender, marital status, family situation and reproductive status such as male, female, transgender, widow(er), mother, etc. and are relevant in domains such as family law, inheritance laws, sexual assault, and violent domestic disputes.
- Caste and Social Stratification Indicators (L_{caste}) includes caste, reservation and hierarchy markers such as SC, ST, OBC, Dalit and Brahmin to aid in categorizing cases related to caste discrimination, reservation and social justice.
- Religious Identity Indicators ($L_{religion}$) include indicators for religion, sectarian and conversion issues (e.g. Hindu, Muslim, Sikh, temple, mosque) which are relevant in communal, personal law and minority rights disputes.
- Economic and Socioeconomic Indicators ($L_{economic}$): poverty, wealth, labor, class (e.g. poor, labourer, farmer, unemployed) as they relate to welfare, bail, sentencing and labour rights.
- Regional, Geographic, and Ethnic Indicators ($L_{regional}$) - regional, tribal, migrant, ethnic markers (tribal, rural, migrant, Gujarati, Adivasi etc.) relevant to land, migration and regional discrimination.
- Disability and Health Vulnerability Indicators ($L_{disability}$). Disability and Health (physical and mental) and also medical vulnerability markers, such as disabled, blind, living with HIV/Aids etc. - relevant for welfare, for need of guardianship, and legal protection.
- Age and Life-Stage Vulnerability Indicators (L_{age}), which include age-related indicators such as minor, juvenile, elderly, or orphan that are relevant to the juvenile justice system, guardianship issues, and elder law.
- Marital and Family Status Indicators (L_{family}) - These Marital and Family Status Indicators together form a rather mixed bag of marital status and family-structure status indicators such as "divorced" or "widow" and also indicate whether the parent of the minor has a female partner and whether the minor lives with single parents; the indicators are relevant for custody matters, hereditary privileges, and domestic law.
- Occupation-Based Social Bias Indicators ($L_{occupation}$) - These Indicators include profession-linked markers (e.g., labourer, sweeper, politician, sex worker) related to occupational prejudice.
- Political and Conflict-Sensitive Identity Indicators ($L_{political}$) comprises of ideological or conflict-related identity indicators (e.g. protestor, extremist, activist) relevant for modelling protest, sedition and conflict litigation scenarios.
- Educational and Literacy Indicators ($L_{education}$) These indicators include literacy and qualifications (such as illiterate, graduate, dropout) that attempt to measure perceived ability or intent.
- Linguistic and Cultural Minority Indicators ($L_{language}$) include language and dialect indicators related to the language or dialect marker (e.g. Hindi-speaking, Urdu-speaking, etc.) that relate to the cultural identity and indicators for minority protection. Tribal dialect indicators also may be included in this category.

A token t_i is included in S if:

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$t_i \in L_{gender} \cup L_{caste} \cup L_{religion} \cup L_{economic} \cup L_{regional} \cup L_{disability} \cup L_{age} \cup L_{family} \cup L_{occupation} \cup L_{disability} \cup L_{plitical} \cup L_{education} \cup L_{language}$

where each L represents a predefined lexical dictionary.

$$BAI = \frac{\sum i \in S|w_i|}{\sum j \in T|w_j|} \quad (9)$$

where:

- T be the set of all tokens in the judgment text
- S ⊂ T be the set of bias-sensitive tokens (caste, gender, religion, economic, regional)
- w_i denotes the attribution weight assigned to token iii through Integrated Gradients (primary), with SHAP used for secondary validation and self-attention used only for contextual encoding.

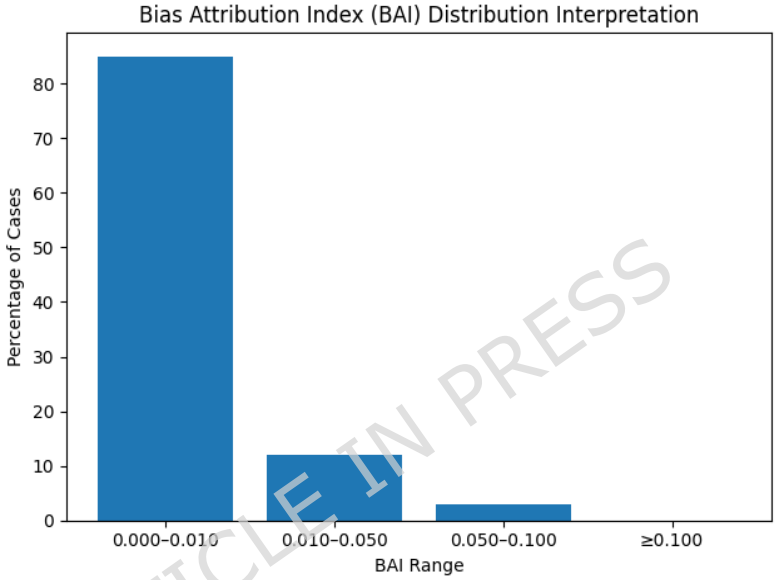


Fig. 3: Distribution of Bias Attribution Index (BAI) Across Evaluated Cases

Table 5. Threshold Interpretation

BAI Range	Interpretation
0.000 ≤ BAI < 0.0100	Negligible bias influence
0.010 ≤ BAI < 0.0500	Mild contextual bias signal
0.050 ≤ BAI < 0.1000	Moderate bias – requires review
BAI ≥ 0.100	Strong bias correlation (unacceptable)

Table 3 presents the threshold interpretation of the Bias Attribution Index (BAI). "In all evaluated probe cases, BAI remained below 0.01, indicating no observable bias influence under the defined attribution criteria*."*

The bias-sensitive token set S contains specific lexical categories which include gendered pronouns and caste and religion identifiers and economic status markers and regional or dialectal terms and legally sensitive demographic references that were developed through domain knowledge and previous legal bias research. The model bases its decisions primarily on evidential and contextual information when BAI values decrease but it relies more on sensitive or non-essential terminology when values increase. The BAI functions to support expert evaluation and interpretive auditing rather than to change predicted outcomes.

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4.4. Error vs Bias Analysis

The research evaluated how errors in classification affected the results based on protected characteristics. The Bias Attribution Index (BAI) remained below 0.01 in all cases where predictions proved incorrect, indicating that the model did not rely on any of the specified characteristics including caste and religion and gender and economic status. The majority of prediction errors occurred when judges faced unclear procedures which led to their discretionary or conditional court decisions. Most minority-class prediction errors were associated with procedurally ambiguous or context-dependent judicial outcomes rather than attribution-sensitive demographic indicators. The results show that most remaining errors stem from legal unclearness and officials' freedom to make decisions instead of detectable algorithmic prejudice which validates the ethical basis of this system. Table 4 summarizes the relationship between prediction errors and bias attribution outcomes.

Table 6. Error vs Bias Matrix

Case Type	Prediction Outcome	Error	BAI	Interpretation
Procedural Housing Case	Other / Procedural	No	<0.01	Correct abstention
Public Employment Dispute	Other / Procedural	No	≈0.00	Legal ambiguity
Criminal Conviction Case	Against Accused	No	0.0023	Evidence-driven
Procedural Borderline Cases (general)	Mixed	Yes	<0.02	Ambiguity-driven, not bias
Minority Verdict Errors	Yes	Yes	<0.03	Data imbalance, not bias

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5. Model Optimization and Empirical Performance Bounds

We conducted specific optimization tests on the NEXAJudicia transformer-based verdict prediction model to determine if the Phase-2 results stemmed from system constraints or natural uncertainty in judicial text information. The research aimed to enhance total prediction accuracy while assessing how the system maintained stability and scalability and maintained ethical standards through various complex modeling scenarios.

5.1 Optimization Strategies

The research investigated four optimization methods which work together to solve three main problems that appear in legal judgment datasets:

Hybrid Late-Fusion Architecture.

The system used a dual-branch neural architecture which applied Legal-BERT-base-uncased transformer encoder to unstructured judgment text in the primary branch and dense layer to process one-hot encoded case types in the secondary branch. The model combined the representations before it used the classification head to obtain additional procedural information.

Cost-Sensitive Learning.

The cross-entropy loss function received inverse class weights because the severe class imbalance made conviction outcomes extremely rare at approximately 7-8% of the total data. The modifications resulted in an improvement in classification

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performance for the "Against Accused" class with recall increasing to 55-60%. This was achieved with little effect on overall classification performance.

Dynamic Text Truncation (Head–Tail Strategy).

The head–tail tokenization method was used to protect important legal information which exists in extended documents by keeping the first factual parts and last decisive sections of court decisions. The model received both evidence stories and court decisions for processing during its inference stage.

Layer Freezing and Regularization.

In order to reduce memorisation, the lower layers of the transformer were selectively frozen and higher dropout rates were used in the classification layer (0.3 to 0.5). This meant the model had to learn more general legal concepts rather than specific examples.

5.2 Empirical Performance Bound

The model achieved its best results through these optimization methods which produced results that matched the Phase-2 results from the beginning. The results from Table 4 show that full fine-tuning of models caused severe overfitting but strong regularization produced underfitting results. The classical baselines showed equal performance which confirmed that the measured results were stable.

5.3 Interpretation: Legal Ambiguity as a Limiting Factor

The procedural ambiguity of judicial decision-making and the strong context dependency of such judgments are reflected in the results obtained by the NEXAJudicia architecture. In a large number of cases within the given dataset, different verdict categories are decided on the basis of a large part of the same factual background information and the same legal arguments. However, from a judicial perspective, such cases are decided differently. It is well known that a wide range of factors that are not recorded in case documents in any form (witness credibility, oral arguments, a judge's procedural discretion, a judge's deliberation process) can lead to different outcomes for cases with identical documented facts and identical documented legal arguments in different instances of the same legal figures.

Unlike image classification in which data is represented as an image, cases in legal domains are represented as text documents. Therefore, accurate judgment prediction models must in addition to high accuracy also extract predictive judicial signals from case text. High model accuracy in this domain could result in models that learn to correlate in context instances of case text with verdicts in a manner that is not related to any legally meaningful reasoning in decision-making. Thus the accuracy in judgment prediction reported for cases in this study as well as in studies reported using similar frameworks represents a number of limitations with currently developed models for judgment prediction from case text.

5.4 Ethical Implications

The NEXA-Judicia system operates at the empirically derived limits of performance. In place of maximising outcomes by complex and aggressive algorithms, the model places emphasis on the model's own reasoning, on a review of built-in bias and on quantifying uncertainty. The system incorporates an attribution-guided analysis and a Bias Attribution Index in order to act as a decision-making tool and a transparency mechanism in the decision process, and not as a judge.

The approach chosen here reflects an awareness of AI systems' capability limitations, in conjunction with legal rules. This awareness safeguards against a false impression of AI certainty that might be given by other systems. Judicial systems using AI must

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incorporate checks and balances so that accountability, audit trails and human supervision are possible.

6 Conclusion and Future Work

6.1 Conclusion

A framework referred to as NEXAJudicia, which provides for the comprehensive prediction of legal judgments, was introduced by the researchers in their study. Its main goals include the assessment of bias and ensuring that the model's decision-making process is transparent. The system provides detailed, attribution-based explanations of both civil and criminal verdicts by incorporating a hybrid transformer-based architecture within a structured two-stage evaluation procedure. A contribution of this research is the formulation of a Bias Attribution Index. This is a statistical measure to gauge how sensitive information affects a model's forecasts. Studies into the system have shown that the method stays within predefined levels of blame, the result of BAI evaluations consistently falling below 0.01 across a number of tests. Further analysis of the results shows that the main issue with predictive outcomes is that they are uncertain, due to the ambiguity of the law and the judges' discretion. This is more of a problem than the current limitations of the algorithm or the capacity of the model itself.

The current standard in legal expert systems suggests that they are not very accurate. This project proves that the observed empirical performance suggests substantial limitations imposed by legal ambiguity and incomplete contextual judicial information by such a computer system is about 81% percent, suggesting that the nature of legal decisions is complex. The framework for responsible judicial decision-making systems involves the employment of attribution-aware validation techniques and systematic bias auditing via A-RAG. This method ensures that judicial systems are transparent and auditable tools for decision-making rather than a means of autonomous judging.

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6.2 Future Work

There are several areas where the proposed framework can be extended. Extending the existing dataset into a more diverse range of legal systems, across different areas of the law and over a period of time for various cases, would facilitate evaluation in different legal systems and varied procedures. Further improvements in alignment with actual legal proceedings could be achieved by incorporating various judicial decisions such as relief that is partial, conditional court orders, and judgments focused on a remedy.

Such work could be undertaken using techniques from temporal modelling. These are statistical methods that analyse how the law changes and evolves with time. This could also involve case law. They would help show how the law's interpretations have changed due to the changing legal framework. In order to carry out longitudinal research, these extensions could be used, assuming they fit in with the reasons why the framework was set up - to ensure that the results are easy to interpret and that a confidence level can be controlled.

A future direction could be to develop training methods that incorporate fairness as an objective, possibly by incorporating a fairness term into the cost function of a machine learning algorithm, or by employing a regularisation technique that encourages the model to be fair. In decisions which carry much weight it is advisable to bring in lawyers to check on the validity of a model’s context, to ensure its ease of understanding and its practicality.

Competing Interests

The authors declare that they have no competing interests.

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Data Availability

The dataset used in this study is publicly available on Kaggle: https://www.kaggle.com/datasets/dayshr/nexajudicia [43].

An archived version of the dataset is available via Zenodo: https://doi.org/10.5281/zenodo.18984839

The dataset is distributed under the Creative Commons Attribution 4.0 International License (CC BY 4.0), which permits unrestricted use, distribution, and reproduction provided that the original authors are properly credited.

Software Availability

The proposed framework does not necessitate a formal software release, as it has been developed primarily for research and experimental validation purposes. Nevertheless, to ensure transparency, reproducibility, and adherence to open science principles, the implementation and associated resources have been made publicly accessible.

The source code of **NEXAJudicia** is available via GitHub: https://github.com/Daysh-Rajabhishek-Ajay-Singh/NEXAJUDICIA.git.

License: MIT

In addition, a citable and archived version of the framework is hosted on Zenodo: https://doi.org/10.5281/zenodo.19162709.

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